

# SHIFTWM: WORLD MODELS THAT MOVE WHAT THEY SAW

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## ABSTRACT

World models let an agent evaluate actions before executing them, and forecasting in the patch-feature space of a frozen visual foundation model keeps them compact. Current latent world models, such as DINO-WM and V-JEPA 2-AC, regenerate the whole feature grid at every future step and feed their predictions back as inputs. Yet in manipulation and surgical scenes most content persists and only moves, so regeneration corrupts the static background and lets errors compound. We introduce SHIFTWM, a latent world model that forecasts by *moving what it has already seen*: for every future step it predicts an action-conditioned soft transport of nearby observed patch features, learned without flow supervision, a gate between moved and kept features and a correction for new content, and it decodes all horizons in parallel. As an output head, it also attaches to published world models. On held-out real-robot video from DROID, Language-Table and the bimanual IWS tasks, and on seven surgical tasks from Open-H, SHIFTWM has the lowest error of all matched output heads, 7.6% below a direct predictor on DROID; as a plug-in, it lowers the error of the 1.3B-parameter V-JEPA 2-AC by 15.7%.

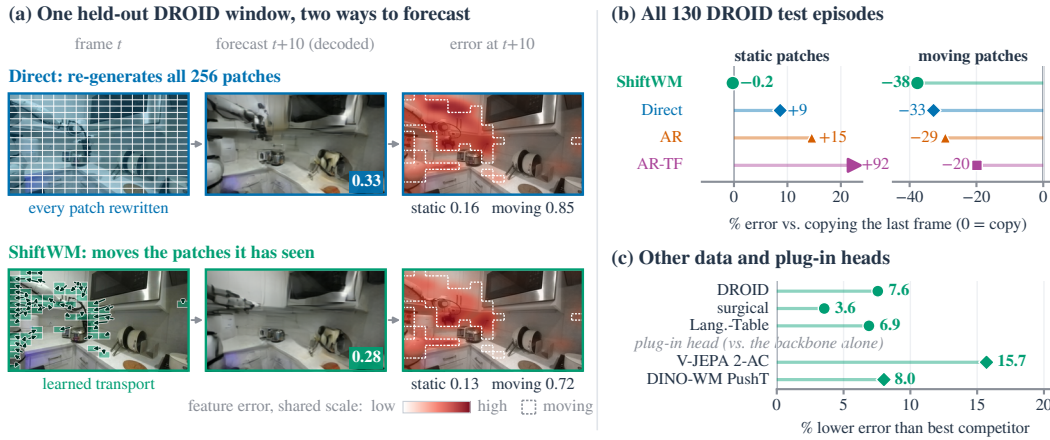


Figure 1: **SHIFTWM moves what it has seen instead of regenerating it.** (a) One held-out DROID window (Appendix D.8); Direct, a matched regression head, rewrites every patch, whereas SHIFTWM moves the arm’s patches (arrows; dashed: the 25% that change most). (b) Error vs. copying the last frame. (c) Selected results: % lower error than the best learned competitor.

## 1 INTRODUCTION

Robots need world models, which predict how observations change in response to actions so that an agent can evaluate candidate actions before executing them (Hafner et al., 2019). Recent latent world models forecast in the patch-feature space of a frozen self-supervised encoder such as DINOv2 or V-JEPA 2, which keeps them compact and lets them plan without generating pixels (Zhou et al., 2025; Assran et al., 2025; Maes et al., 2026b; Sobal et al., 2025). In robot manipulation and surgical

robotics, such a model has to forecast several seconds ahead, respond to the commanded actions and generalise to scenes it has not seen.

Current latent world models regress the complete grid of patch features at every future step, and most feed their own predictions back as inputs. In manipulation and surgical video, however, most of the scene persists, as the table, tissue, tools and robot move but are rarely created or destroyed. Regenerating this persistent content causes three failures. (i) Static background must be rewritten exactly, yet on held-out robot video every regression predictor we train with the same encoder and architecture has higher error on static patches than copying the last frame, a baseline we call *persistence*. (ii) Moving content must be synthesised as a displacement in feature space, a target that grows with the motion and the feature difference between where content comes from and where it goes. (iii) Recursive rollouts compound both errors, and a DINO-WM-style predictor trained with teacher forcing, on true past features instead of its own predictions, has higher error than persistence once it is rolled out. Pixel and feature forecasters exploit motion by warping (Finn et al., 2016; Liu et al., 2017; Šarić et al., 2020), but latent world models for planning do not. On the moving patches of held-out DROID, the quarter with the largest true feature change, picking the observed feature in each patch’s  $7 \times 7$  neighbourhood that is closest to the true future removes 51% of the error of persistence.

We introduce SHIFWWM, a latent world model that forecasts by *moving what it has already seen* (Figures 1 and 2). For every future step, each patch selects where its content comes from among nearby observed features, conditioned on the actions, while a gate keeps static content in place and a small correction adds what is new. SHIFWWM decodes every horizon in parallel from observed features, so its errors never feed back, and because it is an output head, the same move-and-correct forecast attaches to published world models.

We evaluate SHIFWWM against predictors that differ only in the output head, on held-out real-robot and surgical video, attach it to V-JEPA 2-AC and DINO-WM, and test whether the future is in fact a transport of observed content.

In this paper, our major contributions are:

1. **Forecasting by moving, not regenerating.** We introduce SHIFWWM, a latent world model that forecasts each future patch by moving features it has already observed, guided by the actions, instead of regenerating them. Unlike existing latent world models, which regenerate every feature at every step and roll out recursively, it reuses what it has seen and predicts all steps in parallel.
2. **A drop-in head for existing world models.** Added to the 1.3B-parameter V-JEPA 2-AC, it lowers error by 15.7% with only 0.16M extra parameters, and on DINO-WM it lowers error on PushT by 8.0% (Section 5.2).
3. **Lowest error across robots and surgery.** In a matched comparison, SHIFWWM has the lowest error on every benchmark: DROID (including an unseen camera), Language-Table, three bimanual IWS tasks and seven surgical tasks, and it wins on 98% of DROID test episodes (Section 5.1).
4. **The future is mostly the present, moved.** Simply moving observed features removes 51% of the error on moving content. SHIFWWM captures 79% of this gain, and without motion its advantage disappears (Sections 5.3 and 5.4).

## 2 RELATED WORK

**Latent world models.** Predicting dynamics in the embedding space of an encoder rather than in pixels keeps world models small enough to plan with. DINO-WM trains a causal ViT predictor on frozen DINOv2 patch features and plans with model-predictive control, which re-plans at every step (Zhou et al., 2025; Oquab et al., 2024), and V-JEPA 2-AC post-trains an action-conditioned predictor on DROID for zero-shot robot planning (Assran et al., 2025; Khazatsky et al., 2024). PLDM, LeWM and JEPA-WMs learn joint-embedding predictive models from offline trajectories (Sobal et al., 2025; Maes et al., 2026b; Balestrierio & LeCun, 2025; Terver et al., 2026), and DINO-world, DINO-Foresight, VFMF and FlowWM forecast future foundation features, deterministically or generatively (Baldassarre et al., 2025; Karypidis et al., 2025; Boduljak et al., 2025; Porcher et al., 2026). DDP-WM separates sparse dynamic regions from a static background, RLA-WM models residual latent actions (Yin et al., 2026; Zhang et al., 2026a), and recent analyses ask which latent spaces and objectives make predictions action-relevant (Nilaksh et al., 2026; Yeom et al., 2026; Zhang

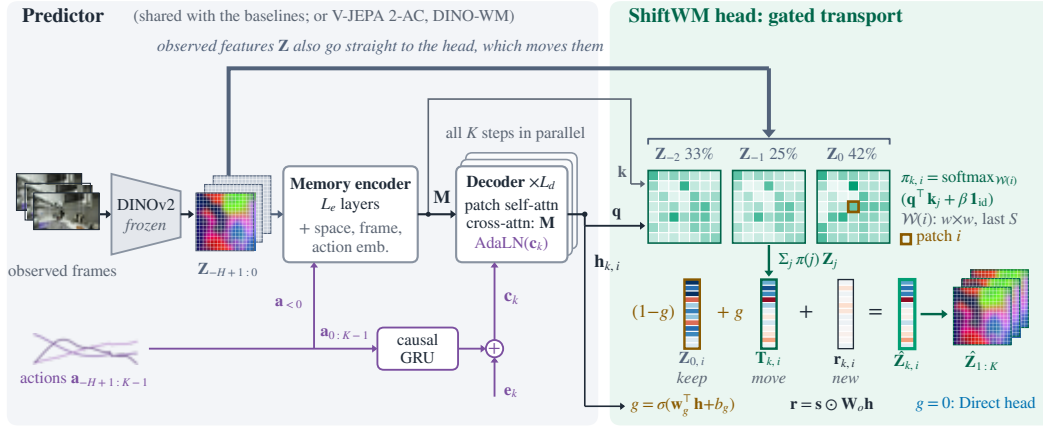


Figure 2: **SHIFTWM architecture.** Left: a predictor, shared with every baseline or taken from a published world model, turns the observed DINOv2 grids and the actions into a query  $\mathbf{h}_{k,i}$  for every patch  $i$  and step  $k$ , all  $K$  steps in parallel. Right: the SHIFTWM head attends over a  $w \times w$  window in the last  $S$  observed grids (Equation (1); weights  $\pi$  from a held-out DROID window,  $k=10$ ), moves the observed features,  $\mathbf{T} = \sum_j \pi(j) \mathbf{Z}_j$ , and outputs  $(1-g)\mathbf{Z}_0 + g\mathbf{T} + \mathbf{r}$  (Equation (2)); with  $g=0$  it is the Direct baseline. Vectors: 12 principal components of one patch.

et al., 2026b; Huang et al., 2026). Pixel-space and model-based reinforcement-learning world models are complementary (Bar et al., 2025; NVIDIA, 2025; Hafner et al., 2025; Hansen et al., 2024). All these predictors synthesise the future feature maps, including content that only moves, whereas SHIFTWM keeps their frozen-encoder interface, forms each forecast as a corrected transport of observed features and attaches to them as a head.

**Multi-step prediction.** Iterating a one-step model compounds its errors (Venkatraman et al., 2015), yet most latent world models are trained for one step and rolled out recursively. DINO-WM uses teacher forcing (Zhou et al., 2025), and V-JEPA 2-AC adds a rollout loss to teacher-forced training (Assran et al., 2025). PlaNet’s latent overshooting reduces the mismatch between one-step training and multi-step use (Hafner et al., 2019), anchored navigation world models resist drift by conditioning on anchor frames (Luan et al., 2026), and decision-oriented objectives align latent predictions with the costs a planner ranks (Wang et al., 2026a). Direct prediction of every horizon avoids feedback, but it still re-synthesises static and moving content, which SHIFTWM keeps or moves instead.

**Motion and feature warping.** Predicting the future by moving existing content has a long history in pixel space. CDNA and DNA transform the previous frame with action-conditioned kernels (Finn et al., 2016), dynamic filter networks generate input-dependent filters (De Brabandere et al., 2016), and Vondrick & Torralba (2017) transform past pixels adversarially. SNA adds skip connections to the first frame (Ebert et al., 2017), SDC-Net and voxel flow resample past pixels along predicted displacements (Reda et al., 2018; Liu et al., 2017), and SAVP and MCVD add stochastic and diffusion models (Lee et al., 2018; Voleti et al., 2022). In feature space, Deep Feature Flow propagates CNN features along optical flow (Zhu et al., 2017), F2F forecasts future features for segmentation (Luc et al., 2017; 2018), F2M warps observed features with a forecast motion field and blends them with a regressed forecast (Šarić et al., 2020), and space-time memory networks copy past features forward by attention (Oh et al., 2019). Recent robot world models reintroduce structure through 3D scene flow (FlowDreamer; Guo et al., 2026), optical-flow velocity (AHEAD; Syed et al., 2026) or object slots (Wu et al., 2023; Mosbach et al., 2025). The pixel methods move pixels, the feature methods move features without actions, and the robot world models use structure to condition a generator rather than as the forecast itself. SHIFTWM carries move-and-correct into latent world models. Unlike F2M, which warps along a single forecast flow field without actions, and space-time memory networks, which read memory by global attention, it learns an action-conditioned soft transport over a local window of several observed frames from the forecasting loss alone. It gates this transport

per patch against an exact identity path, decodes every horizon in parallel and attaches as a plug-in head, and it has lower error than F2M-style warping with optical flow (Section 5.3).

### 3 METHOD

SHIFTWM is a latent world model whose output is a *motion* of the feature grids it has observed. For every future step and every patch, it predicts where in a transport window of observed features the patch’s content comes from, whether the patch stays or moves, and a correction for content that motion cannot explain, as Figure 2 shows with weights from a held-out DROID window. Sections 3.1 and 3.2 define the task and the predictor, Sections 3.3 and 3.4 the transport output and its loss, and Section 3.5 shows why this output suits scenes whose content mostly moves.

#### 3.1 PROBLEM SETTING

A frozen encoder  $E$  maps an image to a grid of  $N = G \times G$  patch features  $\mathbf{Z}_t \in \mathbb{R}^{N \times C}$ . Given  $H$  observed grids  $\mathbf{Z}_{-H+1:0}$ , the actions between them  $\mathbf{a}_{-H+1:-1}$ , and a future action sequence  $\mathbf{a}_{0:K-1}$ , a world model predicts  $\hat{\mathbf{Z}}_{1:K}$ . The forecast  $\hat{\mathbf{Z}}_k$  may depend only on  $\mathbf{a}_{<k}$  (causality). Features are standardised per channel with train-split statistics shared across positions.

#### 3.2 ARCHITECTURE

**Memory encoder.** Each observed patch feature is projected to width  $d$  and receives a spatial embedding, a frame embedding and an embedding of the action applied after that frame.  $L_e$  transformer layers over all  $H \cdot N$  tokens give memory states  $\mathbf{M} \in \mathbb{R}^{H \times N \times d}$ . **Action-prefix queries.** A unidirectional GRU over embedded actions gives a causal prefix state, to which a horizon embedding  $\mathbf{e}_k$  is added to form  $\mathbf{c}_k$ . Query tokens start from the memory states of the last frame, and  $L_d$  decoder layers apply self-attention across patches and cross-attention to  $\mathbf{M}$ , each modulated by  $\mathbf{c}_k$  through adaptive layer normalisation (AdaLN) with zero-initialised gains (Peebles & Xie, 2023). The result is a query  $\mathbf{h}_{k,i} \in \mathbb{R}^d$  for patch  $i$  at step  $k$ , and all  $K$  steps are decoded in parallel.

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#### Algorithm 1 SHIFTWM forecast

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**Input:** grids  $\mathbf{Z}_{-H+1:0}$ , actions  $\mathbf{a}_{-H+1:K-1}$   
**Output:** forecasts  $\hat{\mathbf{Z}}_{1:K}$   
1:  $\mathbf{M} \leftarrow \text{Encoder}(\mathbf{Z}_{-H+1:0}, \mathbf{a}_{<0})$   
2: **for**  $k = 1, \dots, K$  (in parallel) **do**  
3:    $\mathbf{c}_k \leftarrow \text{GRU}(\mathbf{a}_{<k}) + \mathbf{e}_k$   
4:    $\mathbf{h}_k \leftarrow \text{Decoder}(\mathbf{M}, \mathbf{c}_k)$   
5:   **for** each patch  $i$  **do**  
6:      $\pi_{k,i} \leftarrow \text{Eq. (1) over } \mathcal{W}(i)$   
7:      $\mathbf{T}_{k,i} \leftarrow \sum_j \pi_{k,i}(j) \mathbf{Z}_j$   
8:      $g_{k,i} \leftarrow \sigma(\mathbf{w}_g^\top \mathbf{h}_{k,i} + b_g)$   
9:      $\mathbf{r}_{k,i} \leftarrow \mathbf{s} \odot \mathbf{W}_o \mathbf{h}_{k,i}$   
10:     $\hat{\mathbf{Z}}_{k,i} \leftarrow (1 - g_{k,i}) \mathbf{Z}_{0,i} + g_{k,i} \mathbf{T}_{k,i} + \mathbf{r}_{k,i}$   
11:   **end for**  
12: **end for**  
13: **return**  $\hat{\mathbf{Z}}_{1:K}$

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#### 3.3 GATED TRANSPORT

A regression head would write every future feature from  $\mathbf{h}_{k,i}$ , whereas the transport reads it from the observed grids. For query patch  $i$ , let  $\mathcal{W}(i)$  be the  $w \times w$  spatial window around  $i$  in each of the last  $S \leq H$  observed frames. The transport is attention over this window, with a learned bias toward keeping each patch in place:

$$\pi_{k,i}(j) = \text{softmax}_{j \in \mathcal{W}(i)} \left( \frac{1}{\sqrt{d_{\text{key}}}} (\mathbf{W}_{\text{qry}} \mathbf{h}_{k,i})^\top (\mathbf{W}_{\text{key}} \mathbf{M}_j) + \beta \mathbf{1}[j = (0, i)] \right), \quad (1)$$

where  $\mathbf{W}_{\text{qry}}$ ,  $\mathbf{W}_{\text{key}}$  project to key width  $d_{\text{key}}$ ,  $j$  indexes a source patch and frame, and  $\beta$  is a learned identity bias toward the same location in the last frame. The transport moves *observed* features, not latent states,  $\mathbf{T}_{k,i} = \sum_j \pi_{k,i}(j) \mathbf{Z}_j$ . A per-patch gate  $g_{k,i} = \sigma(\mathbf{w}_g^\top \mathbf{h}_{k,i} + b_g)$ , with sigmoid  $\sigma$ , and a correction  $\mathbf{r}_{k,i} = \mathbf{s} \odot \mathbf{W}_o \mathbf{h}_{k,i}$  give

$$\hat{\mathbf{Z}}_{k,i} = (1 - g_{k,i}) \mathbf{Z}_{0,i} + g_{k,i} \mathbf{T}_{k,i} + \mathbf{r}_{k,i}. \quad (2)$$

The first term keeps the last observed feature in place, the second moves observed content, and the third adds what no observed feature contains; with  $g=0$  the output reduces to the Direct baseline  $\mathbf{Z}_0 + \mathbf{r}$ , so the transport term is the only change (Algorithm 1 lists the steps).  $\mathbf{W}_o$  is zero-initialised and  $b_g = -3$ , so an untrained model starts close to persistence, and  $\mathbf{s} \in \mathbb{R}^C$  is a learned per-channel scale.

### 3.4 TRAINING

We minimise the mean squared error over all horizons,  $\mathcal{L}_{\text{pred}} = \frac{1}{K} \sum_k \|\hat{\mathbf{Z}}_k - \mathbf{Z}_k\|^2$ . Anchoring every forecast to observed features makes it easy to ignore the actions, since a forecast that stays close to  $\mathbf{Z}_0$  scores well on average. An optional action-contrastive loss (Equation (10), Appendix C.2) enters only as an ablation, and all forecasting and plug-in results use the prediction loss alone, like every baseline.

### 3.5 MOTION MODEL

We motivate Equation (2) with a simple generative model, drawn for one patch in Figure 3. Suppose that each future patch copies one observed patch, plus zero-mean new content:

$$\mathbf{Z}_k = \mathbf{P}_k \mathbf{Z}_0 + \varepsilon_k, \quad (3)$$

where  $\mathbf{P}_k \in \{0, 1\}^{N \times N}$  is a row-stochastic *motion* matrix (each patch  $i$  copies one source patch  $\rho(i)$  at integer-patch offsets;  $\mathbf{P}_k = \mathbf{I}$  is a static scene; sub-patch motion is covered by soft weights between neighbouring sources) and  $\varepsilon_k$  is new content with  $\mathbb{E}[\varepsilon_k \mid \mathbf{P}_k, \mathbf{Z}_0, \mathbf{a}] = \mathbf{0}$ . Motion is uncertain given the context  $c = (\mathbf{Z}_{\leq 0}, \mathbf{a}_{<k})$ .

**Proposition 1** (The optimal forecast is a transport). *Under Equation (3), the MSE-optimal forecast is  $\mathbb{E}[\mathbf{Z}_k \mid c] = \bar{\mathbf{P}}_k \mathbf{Z}_0$  with  $\bar{\mathbf{P}}_k = \mathbb{E}[\mathbf{P}_k \mid c]$ , a row-stochastic matrix. If the support of each row of  $\mathbf{P}_k$  lies in the transport window  $\mathcal{W}$ , then  $\bar{\mathbf{P}}_k \mathbf{Z}_0$  can be approximated to arbitrary precision by the transport of Equation (1) with  $g=1$  and  $\mathbf{r}=\mathbf{0}$ , given logits that the decoder can express.*

A sufficiently expressive head could reach this optimum as well, but transport parameterises it exactly (①), and the correction  $\mathbf{r}$  absorbs what the model leaves out, such as changes in view-point, lighting and shape. An additive head  $\mathbf{Z}_0 + \mathbf{W}_o \mathbf{h}$  instead has to write out the displacement (②). For a deterministic motion with sources  $\rho(i)$ , its target is  $\mathbf{R} = (\mathbf{P} - \mathbf{I})\mathbf{Z}_0$  with  $\|\mathbf{R}\|_F^2 = \sum_i \|\mathbf{z}_{0,\rho(i)} - \mathbf{z}_{0,i}\|^2$ , the feature difference along the motion, whereas SHIFTWM needs  $\mathbf{r} = \mathbf{0}$  for any such motion within the window. This concerns what each head must represent, not what it can learn. On static patches the identity path is exact, whereas a regression head must output exactly zero. Finally, a recursive predictor that is  $L$ -Lipschitz with one-step error at most  $\delta$  has a  $k$ -step error bounded by  $\delta \sum_{j=0}^{k-1} L^j$ , which grows geometrically for  $L > 1$  (Venkatraman et al., 2015) (③). SHIFTWM decodes each horizon from observed features, so its errors never feed back. Appendix B.2 gives the derivations.

**Testable predictions.** Each claim predicts a result that we check. If the future is a transport, choosing among the episode’s own observed features should remove most of the error on moving content, and features from an unrelated episode should not (oracle and null control, Section 5.3). If the regression target grows with the motion, the gain should concentrate on moving patches while static patches are copied (moving versus static split, Section 5.3). If recursion compounds errors, the gain over AR should grow with the horizon (Section 5.1).

## 4 EXPERIMENTS

We first compare SHIFTWM with predictors that differ only in the output head on held-out robot and surgical video (Section 5.1), then attach its head to two published world models (Section 5.2), test whether its gains come from moving observed content (Section 5.3), and remove one component at a time (Section 5.4). Model selection uses validation splits only, and every test split is scored once.

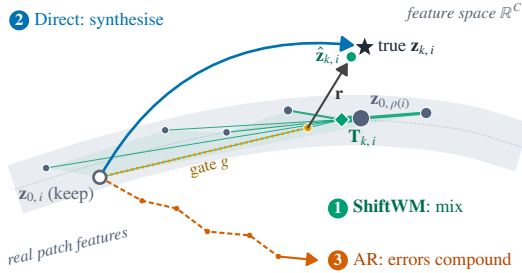


Figure 3: **How each head reaches one patch’s future feature** (circle: last observed; star: true future). ① SHIFTWM mixes nearby observed features (line width =  $\pi$ ) into  $\mathbf{T}$ , gated, plus a small correction  $\mathbf{r}$ . ② A direct additive head synthesises the whole displacement. ③ An autoregressive (AR) head feeds its errors forward.

Table 1: **Held-out forecasting.** Left: feature MSE  $\downarrow$  per benchmark (mean over horizons unless noted; training seeds: DROID 3, Open-H 1, Language-Table 1, IWS 1). Right: on DROID, error on the 25% of patches that change most (moving) and on the rest (static), and action-ranking accuracy, the share of windows where the true actions give a lower error than another episode’s. **Best, second**; †: paired 95% bootstrap CI (episodes) vs. the best baseline excludes 0; –: not run or not applicable.

| Method                            | forecast error by benchmark |               |               |               |               |               | DROID analysis |              |                  |
|-----------------------------------|-----------------------------|---------------|---------------|---------------|---------------|---------------|----------------|--------------|------------------|
|                                   | DROID                       |               | DROID cam 2   | Open-H        | Language-     | IWS           | moving         | static       | action           |
|                                   | avg                         | $k=10$        | (zero-shot)   | surgical      | Table         | bimanual      | $\downarrow$   | $\downarrow$ | rank. $\uparrow$ |
| Persistence                       | 0.249                       | 0.326         | 0.285         | 0.444         | 0.365         | 0.741         | 0.726          | 0.090        | –                |
| Linear extrapolation              | 4.812                       | 11.855        | 5.297         | 10.824        | 8.437         | –             | –              | –            | –                |
| AR-TF (DINO-WM-style)             | 0.275                       | 0.417         | 0.314         | 0.285         | –             | 0.264         | 0.582          | 0.173        | 84.9             |
| AR (rollout-trained)              | 0.206                       | 0.264         | 0.244         | 0.235         | 0.122         | 0.233         | 0.513          | 0.103        | 81.0             |
| Direct (cross-attn)               | 0.196                       | 0.250         | 0.233         | 0.234         | 0.120         | 0.233         | 0.488          | 0.098        | 89.8             |
| SHIFTWM (ours)                    | <b>0.181†</b>               | <b>0.231†</b> | <b>0.215†</b> | <b>0.226†</b> | <b>0.111†</b> | <b>0.225†</b> | <b>0.453</b>   | <b>0.090</b> | <b>92.0</b>      |
| error reduction vs. best baseline | <b>+7.6%</b>                | <b>+7.3%</b>  | <b>+7.5%</b>  | <b>+3.6%</b>  | <b>+6.9%</b>  | <b>+3.4%</b>  | <b>+7.2%</b>   | <b>+8.2%</b> | <b>+2.2 pt</b>   |

**Datasets.** **DROID** (Khazatsky et al., 2024) contains episodes (recorded task demonstrations) of a real Franka robot arm, split by recording session. We train on the primary exterior camera and test zero-shot transfer to the second; one step is 0.33 s, its 35-D action block holds the five commanded gripper poses and openings of that step, and we forecast  $K=10$  steps (3.3 s). **Open-H Hamlyn** (Open-H-Embodiment Consortium, 2026) contains seven surgical training tasks, two on table-top phantoms (artificial practice models) and five on ex-vivo porcine tissue, recorded with the dVRK (da Vinci Research Kit) robot, 80-D action blocks for its two arms and episode-disjoint splits per task. **Language-Table** (Lynch et al., 2023) shows an xArm robot arm pushing rigid blocks (a fixed Open-X subset; Open X-Embodiment Collaboration, 2024), and **IWS** (Wang et al., 2026b; Zhang et al., 2026a) records real bimanual arms pushing a T-block, handling a box and manipulating a rope, scored on the official validation handles, which come from recordings disjoint from the training recordings that also supply our checkpoint-selection split. For the plug-in study, we use the simulated PushT (pushing a T-shaped block to a target pose) and Wall (moving through a door in a wall) tasks with the official **DINO-WM** code.<sup>1</sup> Table 3 lists sizes, splits and licences, and Figure 9 maps all benchmarks, predictors and evaluations and defines the terms.

**Baselines.** All learned predictors share a frozen DINOv2-S/14 encoder<sup>2</sup> ( $16 \times 16$  patches, no pooling), the same memory encoder and decoder (width 384,  $L_e=L_d=4$ ,  $\approx 22$ M parameters), optimiser, schedule, steps and batch size, and differ only in the output head and rollout. **Direct** replaces Equation (2) with  $\mathbf{Z}_0 + \mathbf{W}_o \mathbf{h}_k$ ; it has the same cross-patch attention as SHIFTWM, so it controls for spatial communication. **AR** predicts a residual to the most recent grid and is rolled out recursively, with gradients through the rollout. **AR-TF** follows DINO-WM, as it is trained with teacher forcing to predict full next-step features and rolled out at test time. **Persistence** copies  $\mathbf{Z}_0$ , and **Linear** extrapolates the last feature difference, which multiplies frame-to-frame feature noise by up to  $K$  and is therefore far worse than persistence. As published models, we use V-JEPA 2-AC<sup>3</sup> and DINO-WM with and without the SHIFTWM head.

**Metrics and statistics.** The primary metric is standardised feature MSE on held-out test windows of  $H=3$  observed and  $K=10$  future steps ( $K=12$  on IWS), averaged over horizons  $k$  (steps ahead) and at the final step. We also report *skill*, the fraction of persistence error a model removes, which is scale-free and therefore compares models in different feature spaces. Episodes (recording sessions for DROID) are the statistical unit, and Appendix C gives all further details.

## 5 RESULTS AND DISCUSSION

### 5.1 HELD-OUT FORECASTING

Table 1 compares SHIFTWM with predictors that share its frozen encoder and architecture and differ only in how the forecast is formed. On held-out DROID, SHIFTWM has the lowest error of all methods, 7.6% below Direct, 12.2% below the rollout-trained AR and 34.3% below the DINO-WM-style AR-TF, which predicts full features rather than changes and is worse than persistence

<sup>1</sup> [https://github.com/gaoyuezhou/dino\\_wm](https://github.com/gaoyuezhou/dino_wm)

<sup>2</sup> <https://github.com/facebookresearch/dinov2>

<sup>3</sup> <https://github.com/facebookresearch/vjepa2>

Abhijit (AI Assisted): the DINO-WM-style AR-TF is missing here ('-' in Table 1). run it or say why. [mdw]

from  $k=5$  on. It removes 27.5% of the persistence error, against 21.6% for Direct and 17.4% for AR. **The gains transfer.** Zero-shot on DROID’s second exterior camera, which no model saw in training, SHIFTWM is 7.5% below Direct and 11.8% below AR. On Language-Table, where a robot pushes rigid blocks, it is 6.9% and 8.6% below them, and on the seven Open-H surgical tasks 3.6% and 4.0%. On the three real bimanual IWS tasks it is 3.5% and 3.4% below them, with the lowest average error on each task, significantly on Box and Rope (Table 5). Paired 95% intervals of the gain over the best baseline are [6.8, 8.3]% on DROID (resampling 57 test sessions), [6.9, 8.1]% on its second camera, [6.4, 7.4]% on Language-Table, [3.3, 3.9]% on Open-H and [3.0, 3.8]% on IWS.

**The gains hold across horizons and surgical tasks.** The gain over AR-TF grows from 10% at  $k=1$  to 45% at  $k=10$  on DROID as its errors compound (Section 3.5), and the gain over Direct is positive at every horizon on every dataset, that over AR everywhere except at long IWS horizons (PushT from  $k=6$ , Box from  $k=10$ ). SHIFTWM also ranks actions best on DROID (92.0% against 89.8% for Direct; Table 1, right). Appendix D.1 gives the per-horizon, per-task and action-ranking results.

## 5.2 PLUG-IN HEADS FOR PUBLISHED MODELS

Because SHIFTWM is an output head, we attach it to two published models without other changes. The head blends the backbone’s prediction  $\mathbf{B}$  with transported observed features,  $\hat{\mathbf{Z}} = (1 - g)\mathbf{B} + g\mathbf{T}$ , with the gate initialised near zero so that training starts from the unmodified model. Each pair shares data, schedule, loss and seed.

**V-JEPA 2-AC.** The released model (Assran et al., 2025) (ViT-g encoder, 300M predictor, trained on DROID) removes 4.1% of persistence error zero-shot on our test windows, 24.6% after fine-tuning, and 36.4% after fine-tuning *with* the head under the identical budget (Figure 4). The head thus lowers error by 15.7%, and by 16.9% on moving patches, for 0.16M added parameters. The gate opens from 1.8% to 56% at the selected checkpoint and 65% by the end of fine-tuning, and validation error is lower at every evaluation for both seeds, as Appendix D.3 shows. The model’s training data likely overlap our test trajectories, which affects all three rows, so the head’s gain is relative to the same starting point, data and budget.

**DINO-WM.** With the official code (Zhou et al., 2025), the head lowers teacher-forced latent error, the one-step feature error given true past features, by 8.0% on PushT. On Wall it raises latent error by 18.4%, as DINO-WM conditions on a single history frame ( $S=1$ ), which leaves the head one observed frame to move and, after the first rollout step, only DINO-WM’s own predictions.

## 5.3 WHY TRANSPORT HELPS

**Where do the gains come from?** The regression target of Section 3.5 predicts gains on moving content and exact copies of static content. We split every test window into *moving* patches, the 25% with the largest true feature change (a property of the data shared by all methods), and *static* patches. On moving patches, SHIFTWM has 7.2% lower error than Direct and 11.8% lower than AR (Table 1, right). On static patches, every other learned predictor is *worse than copying*, as regenerating the grid corrupts background that should not change, whereas SHIFTWM is the only learned model at or below copy. Across all ten deciles of true motion, SHIFTWM has lower error than Direct (+5.1% to +13.2%), with the gains concentrated on the arm and gripper (Figure 5a,c). Per episode (seed 0), SHIFTWM has lower error than Direct on 98% and than AR on 98% of the 130 DROID test episodes long enough for one window (two of 132 are shorter than 13 steps), and its few losses all lie in the less-moving half (Figure 6); the absolute gain grows with motion even as the relative gain shrinks (Appendix D.6).

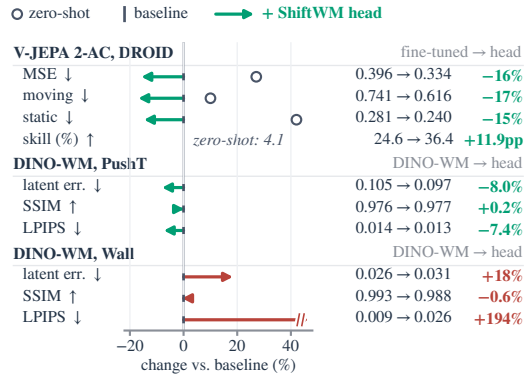


Figure 4: **SHIFTWM as a plug-in head:** change from adding the head to the same model (same data, schedule, loss). V-JEPA 2-AC: DROID test, base = fine-tuned (2 seeds), circles = zero-shot; DINO-WM: validation. Green better, red worse; numbers in Table 2.

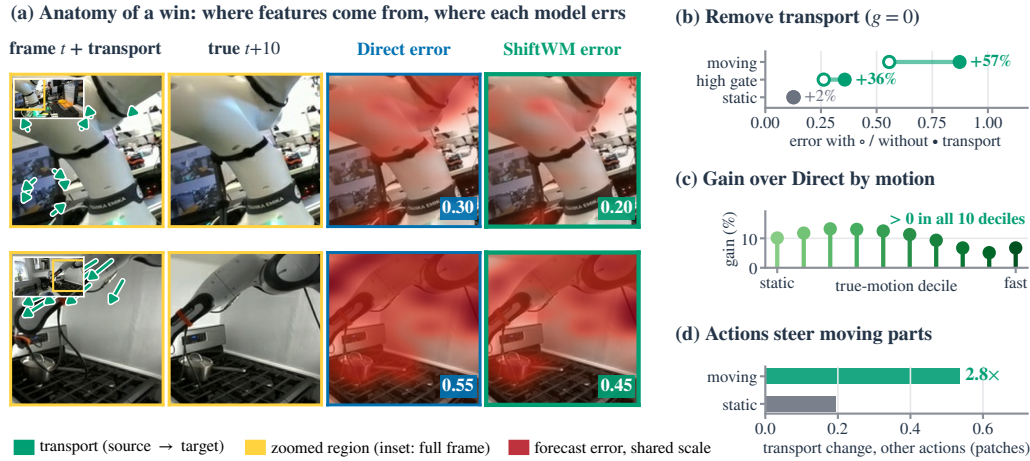


Figure 5: **Where and why SHIFTWM wins (held-out DROID,  $k=10$ ).** (a) Transport arrows, true future, and Direct vs. SHIFTWM error on the same zoom (selected windows). (b) Removing transport. (c) Gain per motion decile. (d) Action steering.

**Reliance on transport.** The controlled evidence is the retrained Direct head, identical except for transport, over which SHIFTWM lowers moving-patch error by 7.2%. At test time, setting the gate to zero, so that each forecast becomes  $\mathbf{Z}_0 + \mathbf{r}$ , raises the error by 56.8% on moving patches and by 35.9% on patches with a high learned gate, but only by 1.8% on static patches (Figure 5b). The trained model therefore moves content through the transport, not through the correction. Swapping in another episode’s future actions changes the transport field, the expected source offset of each patch, 2.8 $\times$  more on moving than on static patches (Figure 5d), so the actions steer *where* features are moved.

**Is the future a transport?** As a model-free check of Proposition 1, we pick, for each moving patch of 3,923 DROID test windows, the observed feature in its transport window closest to the true future, an *oracle move*. At  $k=10$  the oracle move cuts the error of persistence by 51%, so half of the future is already in the observed grid, displaced; the same oracle with candidates from an unrelated test episode is 40% worse than persistence (18% worse even from the same lab), so the gain comes from the episode’s own content, not from choosing among many candidates. SHIFTWM recovers 79% of this gain (Direct 71%, AR 69%); as Direct differs only in the transport, the gap between the shares is what transport adds. SHIFTWM is better than Direct on 64% of moving patches, and has lower error than Direct and AR in all 100 cells of 10 horizons by 10 motion deciles, as Appendix D.6 details.

**Beyond feature error.** The learned transport is more than flow warping, as SHIFTWM has 25.4% lower error than warping  $\mathbf{Z}_0$  with RAFT optical flow<sup>4</sup> (F2M-style) and 18.1% lower than warping with the true future flow, 18% even without the correction term. Labelling each forecast patch by its most similar observed patch (cosine) under the SAM 2.1 arm mask of the last observed frame, without any decoder, SHIFTWM has the highest arm IoU on moving DROID windows, +2.0 points above Direct and +4.2 above AR, with paired intervals that exclude zero; Appendix D.7 gives both protocols.

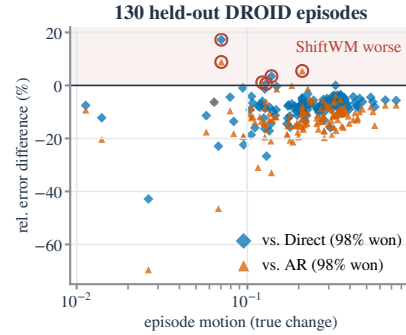


Figure 6: **Per-episode gain** (DROID test, seed 0, mean over  $k$ ; below 0: SHIFTWM better; circles: losses).

<sup>4</sup> torchvision implementation, <https://github.com/pytorch/vision>.

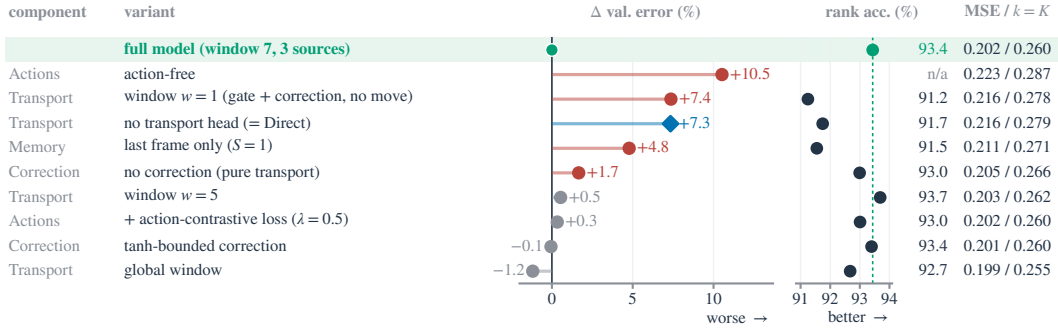


Figure 7: **Ablations on DROID validation** (seed 0; selection on validation only), sorted by  $\Delta$ , the change of validation MSE against the full model (positive = worse); action-ranking accuracy and absolute MSE on the right. The full model’s seed range is 0.2%, so gaps below about 0.3% are within seed noise.

**Statistical analysis.** All intervals are paired bootstraps of per-episode differences (sessions for DROID); every dagger in Table 1 survives Holm correction, and with one training seed (Open-H, Language-Table, IWS) intervals reflect episode variance only (Appendix C.3).

#### 5.4 ABLATIONS

Figure 7 changes one component at a time on DROID, scored on validation. Replacing the transport head by a direct regression head (Direct) raises validation error by 7.3% and lowers action-ranking accuracy, and restricting transport to the last observed frame ( $S=1$ ) raises it by 4.8%. **The gain comes from moving content.** A  $1 \times 1$  transport window ( $w=1$ ) keeps the gate and correction but cannot move content, and its error rises by +7.4%, as much as for Direct. A  $5 \times 5$  window is within 0.5%, and a global window lowers error slightly (-1.2%) but ranks actions less well (92.7 vs. 93.4%) and scores 768 instead of 147 candidates per patch, so the local window keeps the benefit of transport with a fifth of the candidates. A tanh-bounded correction matches the learned scale (-0.1%), and removing the correction costs +1.7%. Removing the actions costs most (+10.5%), so the forecasts depend on the commanded motion and not only on the observed frames, and an action-contrastive loss adds nothing (+0.3%), as the transport already uses the actions.

## 6 CONCLUSION

SHIFTWM forecasts by moving what it has already seen and correcting only what is new. Against matched predictors, it has the lowest held-out error on DROID (including an unseen camera), Language-Table, three bimanual IWS tasks and seven surgical tasks, and it wins on 98% of DROID test episodes. As a plug-in head, it lowers the error of the 1.3B-parameter V-JEPA 2-AC by 15.7% with 0.16M added parameters. Its transport, gate and correction also make explicit where each forecast feature comes from. Move-and-correct is a simple and general output parameterisation, and we encourage latent world models to form their forecasts from what they have observed.

**Limitations.** As a plug-in to DINO-WM on Wall, whose single history frame leaves the head little observed content to move, SHIFTWM raises latent error by 18.4%; consistently, restricting SHIFTWM to one source frame raises its DROID validation error by 4.8%. Content never observed relies on the correction, and the forecasts are deterministic, which stochastic feature forecasters could complement. Lower forecast error does not by itself raise PushT planning success (Appendix D.4). Language-Table test and training segments may come from one long unscripted recording, and the surgical data are external-camera views of practice tasks, not patient endoscopy.

#### REPRODUCIBILITY STATEMENT

Code, configurations, per-episode evaluation records and trained checkpoints will be released; the model is also packaged as an installable library (Appendix B.4). Appendix B.1 lists every hyper-

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parameter. Dataset splits are deterministic and documented in Appendix C.1. All datasets are public, namely DROID (CC-BY-4.0), Open-H-Embodiment (CC-BY-4.0), Language-Table via Open-X (CC-BY-4.0), the IWS recordings of the RLA-WM release (no licence stated; used for research as released), the LeWM planning datasets (MIT) and the DINO-WM releases.

#### ETHICS STATEMENT

The surgical data are public recordings of robotic procedures on table-top phantoms and ex-vivo porcine tissue, released under CC-BY-4.0. We make no clinical claims, and the models are not intended for clinical use.

#### AI USE STATEMENT

In this work, we used generative AI tools to assist with writing and debugging research code, orchestrating experiments, surveying related work, and editing the manuscript text. All AI-assisted code was tested, and every reported number is produced by logged runs that the authors reviewed. Bibliographic entries were verified against the original sources. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

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**Appendix overview.** The appendix reads as a complete paper, with the background (Appendix A), the extended method (Appendix B), experimental details (Appendix C), full results and analyses including closed-loop planning (Appendix D), and a discussion (Appendix E).

## A BACKGROUND

### A.1 THE TASK AND ITS USES

An action-conditioned latent world model receives the last  $H$  observed frames of a scene, encoded by a frozen visual foundation model into grids of patch features, the actions applied between them, and a candidate sequence of  $K$  future actions. It predicts the patch-feature grids of the next  $K$  frames without seeing them, and it never generates pixels, as the forecast lives in the feature space that downstream consumers read. Such forecasts serve (i) planning, where a model-predictive planner forecasts many candidate action sequences and executes the one whose forecast is closest to a goal (Zhou et al., 2025; Maes et al., 2026b), (ii) policy evaluation, where a world model scores a policy’s proposed actions before a real robot executes them, and (iii) anticipation, where a surgical system forecasts where instruments and tissue will be a few seconds ahead for assistance and safety monitoring. All three need forecasts that are accurate many steps ahead, respond to the actions and hold on unseen scenes.

Figure 8 shows the task on two held-out episodes, a Franka arm in a real kitchen (DROID) and a dVRK instrument over a practice phantom (Open-H Hamlyn), chosen as the test episodes with the largest feature change. The model receives three frames encoded into  $16 \times 16$  grids of 384-D DINOv2 features, the actions between them and ten future action blocks, and it predicts the feature grids 3.3 s ahead. Change concentrates on the robot and the objects it touches, as in these two windows a quarter of the patches carry more than 40% of the total change. Most future features are thus already in the observed grid, displaced by motion, and only a small part is new content.

### A.2 EVALUATION OVERVIEW AND TERMS

Figure 9 maps every benchmark, the evaluation axes it serves and the predictors evaluated on it, and the list under it defines the remaining terms.

### A.3 FURTHER RELATED WORK

Pixel-space generative world models learn action-controllable video simulators, from navigation (NWM; Bar et al., 2025) to foundation models for physical AI (Cosmos; NVIDIA, 2025) and interactive environments (Genie; Bruce et al., 2024), and model-based reinforcement learning learns task-specific latent dynamics jointly with rewards and policies (Hafner et al., 2019; 2025; Hansen et al., 2024). These systems optimise sample fidelity or online return with their own encoders and data, so they do not share our setting of a frozen encoder with targets in common feature coordinates, and we treat them as complementary. Surgical world modelling is likewise driven by generative video models, such as Endora for endoscopy video (Li et al., 2024), SurgWM with latent-action control (Koju et al., 2025) and Cosmos-H-Surgical for policy learning (He et al., 2025). We

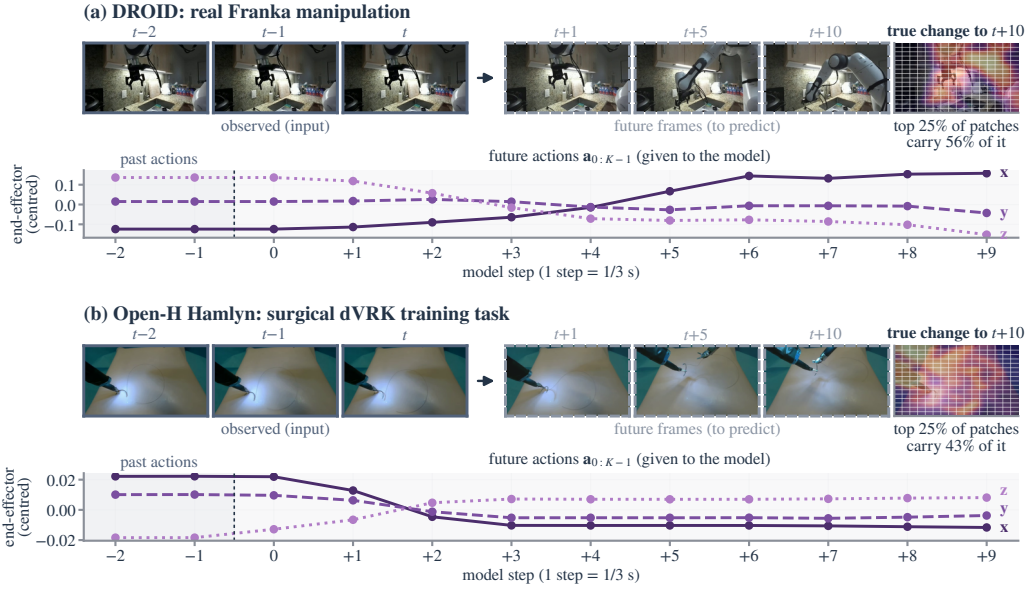
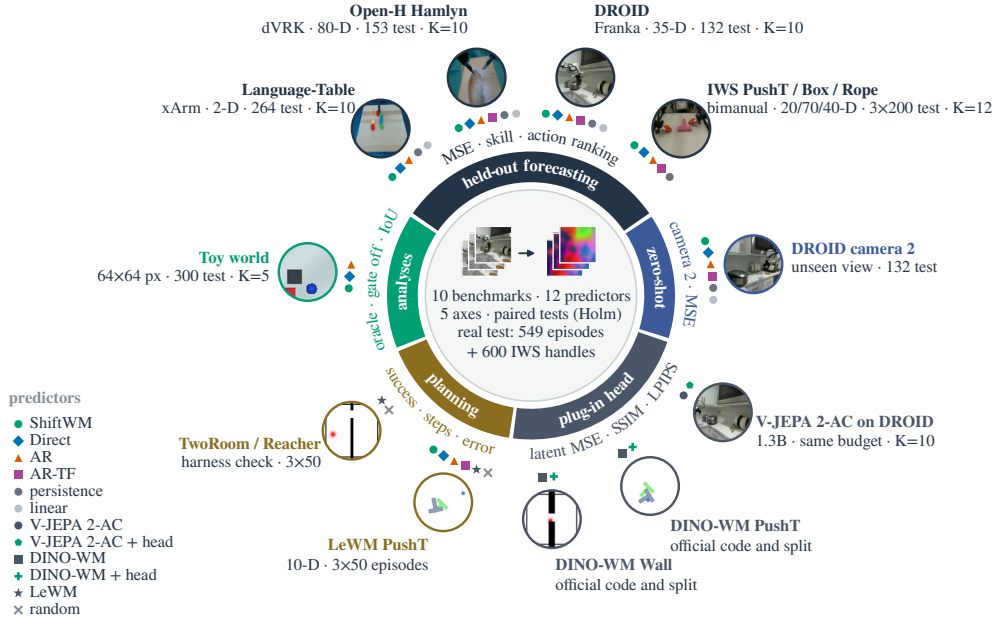


Figure 8: **The forecasting task on held-out episodes.** Three observed frames and the future actions are given, and the future feature grids are predicted. Right: true feature change over the  $16 \times 16$  grid.

instead use Open-H dVRK recordings (Open-H-Embodiment Consortium, 2026) to test kinematics-conditioned forecasting in feature space, and for control we follow the latent planning evaluation of stable-worldmodel (Maes et al., 2026a).



#### Robot data

|                          |                                                                |
|--------------------------|----------------------------------------------------------------|
| <b>Episode / session</b> | one demonstration; episodes recorded consecutively at one site |
| <b>End-effector</b>      | the tool at the arm's tip, a two-finger gripper                |
| <b>Action block</b>      | all commands of one model step, concatenated                   |
| <b>Surgical data</b>     |                                                                |
| <b>Practice task</b>     | a standard surgeon exercise, e.g. peg transfer                 |
| <b>Phantom</b>           | an artificial practice model of tissue                         |
| <b>dVRK</b>              | da Vinci Research Kit, a surgical robot                        |
| <b>Scene camera</b>      | an external camera, not an endoscope                           |
| <b>Planning</b>          |                                                                |
| <b>MPC / CEM</b>         | plan, act, re-plan; sample and refit                           |
| <b>Baselines</b>         |                                                                |
| <b>Direct</b>            | regresses each future grid as $\mathbf{Z}_0$ + residual        |
| <b>AR</b>                | residual to the latest grid, rolled out recursively            |
| <b>AR-TF</b>             | DINO-WM-style: teacher-forced next step, rolled out            |

#### ShiftWM

|                           |                                                                     |
|---------------------------|---------------------------------------------------------------------|
| <b>Patch features</b>     | frozen DINOv2 $16 \times 16$ grid                                   |
| <b>Transport window</b>   | $w \times w$ patches in the last $S$ frames                         |
| <b>Transport</b>          | content moved from observed patches                                 |
| <b>Correction</b>         | added residual for new content                                      |
| <b>Gate</b>               | per-patch weight: keep vs. move                                     |
| <b>Oracle move</b>        | observed feature closest to the truth                               |
| <b>Evaluation</b>         |                                                                     |
| <b>Persistence</b>        | copy the last observed grid                                         |
| <b>Rollout</b>            | own predictions fed back as inputs                                  |
| <b>Moving patches</b>     | 25% with the largest true change                                    |
| <b>Teacher forcing</b>    | train on true past, not own output                                  |
| <b>Skill</b>              | share of persistence error removed                                  |
| <b>Session split</b>      | no recording session in two splits                                  |
| <b>Other</b>              |                                                                     |
| <b>Window, horizon</b>    | $H=3$ observed and $K=10$ future steps; $k$ steps ahead             |
| <b>Open / closed loop</b> | forecasting without new observations / re-planning after each chunk |
| <b>Success</b>            | the final state is within the task's tolerance of the goal          |
| <b>Proprioception</b>     | the robot's measured joint and gripper state                        |
| <b>Ex vivo</b>            | tissue removed from the body                                        |

Figure 9: **Evaluation overview.** Evaluation axes (inner ring, with metrics), benchmarks (real frames) and the predictors evaluated on each (markers); scope only, results are in Section 5 and Appendix D. Below: definitions of the other terms.

## B EXTENDED METHOD

### B.1 ARCHITECTURE

The memory encoder attends over all  $H \cdot N=768$  tokens of the three observed grids, each query-decoder layer applies self-attention, cross-attention to memory and an MLP under AdaLN-zero, and actions are embedded by a 2-layer MLP and a single-layer GRU of width  $d$ . The transport uses key dimension 64 and a  $7 \times 7$  window over  $S=3$  source frames (147 candidates per query), and the correction head applies LayerNorm, a zero-initialised linear map to  $C$  and a per-channel scale initialised at 1. Table 4 lists all sizes and settings.

## B.2 PROOFS

Throughout,  $c = (\mathbf{Z}_{\leq 0}, \mathbf{a}_{<k})$  is the context,  $\mathbf{e}_j$  the  $j$ -th standard basis vector, and  $\mathbf{z}_{0,j}$  the  $j$ -th row of  $\mathbf{Z}_0$ .

*Proof of Proposition 1. Optimal forecast.* Under squared error the optimal forecast is the conditional mean. Since  $\mathbf{Z}_0$  is determined by  $c$  and the new content has zero conditional mean, the tower property gives

$$\mathbb{E}[\mathbf{Z}_k | c] = \mathbb{E}[\mathbf{P}_k | c] \mathbf{Z}_0 + \mathbb{E}[\mathbb{E}[\boldsymbol{\varepsilon}_k | \mathbf{P}_k, c] | c] = \bar{\mathbf{P}}_k \mathbf{Z}_0. \quad (4)$$

*Structure of  $\bar{\mathbf{P}}_k$ .* Row  $i$  of  $\mathbf{P}_k$  is  $\mathbf{e}_{\rho(i)}^\top$ , so row  $i$  of  $\bar{\mathbf{P}}_k$  is  $\sum_j \Pr[\rho(i) = j | c] \mathbf{e}_j^\top$ , a probability vector whose support lies in  $\mathcal{W}(i)$  whenever the support of  $\rho(i)$  does.

*Realisation by transport.* With  $g = 1$  and  $\mathbf{r} = \mathbf{0}$ , Equation (2) gives  $\hat{\mathbf{z}}_{k,i} = \sum_{j \in \mathcal{W}(i)} \pi_{k,i}(j) \mathbf{z}_{0,j}$ . Let  $\mathcal{S}_i$  be the support of row  $i$  of  $\bar{\mathbf{P}}_k$ ,  $u = |\mathcal{W}(i) \setminus \mathcal{S}_i|$ , and choose logits  $\ell_j = \log \bar{P}_{k,ij}$  for  $j \in \mathcal{S}_i$  and  $\ell_j = -\Lambda$  otherwise. Since  $\sum_{j \in \mathcal{S}_i} \bar{P}_{k,ij} = 1$ ,

$$\pi_{k,i}(j) = \frac{\bar{P}_{k,ij}}{1 + ue^{-\Lambda}} \quad (j \in \mathcal{S}_i), \quad \pi_{k,i}(j) = \frac{e^{-\Lambda}}{1 + ue^{-\Lambda}} \quad (j \notin \mathcal{S}_i), \quad (5)$$

so the total-variation distance to row  $i$  of  $\bar{\mathbf{P}}_k$  is  $2ue^{-\Lambda}/(1 + ue^{-\Lambda}) \leq 2ue^{-\Lambda}$  and

$$\|\hat{\mathbf{z}}_{k,i} - (\bar{\mathbf{P}}_k \mathbf{Z}_0)_i\| \leq 2ue^{-\Lambda} \max_{j \in \mathcal{W}(i)} \|\mathbf{z}_{0,j}\| \xrightarrow{\Lambda \rightarrow \infty} 0. \quad (6)$$

With  $S > 1$  source frames the same argument holds with  $\mathcal{W}(i)$  ranging over patches of all observed frames.  $\square$

*Regression target of Section 3.5.* An additive head outputs  $\mathbf{Z}_0 + \mathbf{R}$ . Matching  $\mathbf{P}\mathbf{Z}_0$  requires

$$\mathbf{R} = (\mathbf{P} - \mathbf{I})\mathbf{Z}_0, \quad \mathbf{R}_i = \mathbf{z}_{0,\rho(i)} - \mathbf{z}_{0,i}, \quad \|\mathbf{R}\|_F^2 = \sum_i \|\mathbf{z}_{0,\rho(i)} - \mathbf{z}_{0,i}\|^2. \quad (7)$$

For SHIFTW, take the one-hot transport  $\pi_{k,i} = \mathbf{e}_{\rho(i)}$  (the limit  $\Lambda \rightarrow \infty$  above, admissible since  $\rho(i) \in \mathcal{W}(i)$ ) and  $g = 1$ . Then  $\hat{\mathbf{z}}_{k,i} = \mathbf{z}_{0,\rho(i)} = (\mathbf{P}\mathbf{Z}_0)_i$  for every  $i$ , so the required correction is  $\mathbf{r} = \mathbf{0}$ .  $\square$

*Compounding bound of Section 3.5.* Let the recursive rollout be  $\hat{\mathbf{Z}}_j = f(\hat{\mathbf{Z}}_{j-1})$  with  $\hat{\mathbf{Z}}_0 = \mathbf{Z}_0$ , the truth  $\mathbf{Z}_j = f(\mathbf{Z}_{j-1}) + \boldsymbol{\eta}_j$  with  $\|\boldsymbol{\eta}_j\| \leq \delta$ , and  $e_j = \|\hat{\mathbf{Z}}_j - \mathbf{Z}_j\|$ . By the triangle inequality and the Lipschitz property,

$$e_j \leq \|f(\hat{\mathbf{Z}}_{j-1}) - f(\mathbf{Z}_{j-1})\| + \|\boldsymbol{\eta}_j\| \leq L e_{j-1} + \delta, \quad e_0 = 0. \quad (8)$$

By induction on  $k$  (the case  $k = 1$  is  $e_1 \leq \delta$ ; if  $e_{k-1} \leq \delta \sum_{j=0}^{k-2} L^j$  then  $e_k \leq L\delta \sum_{j=0}^{k-2} L^j + \delta$ ),

$$e_k \leq \delta \sum_{j=0}^{k-1} L^j = \begin{cases} \delta \frac{L^k - 1}{L - 1}, & L \neq 1, \\ k\delta, & L = 1, \end{cases} \quad (9)$$

which grows geometrically for  $L > 1$ . A direct predictor forms  $\hat{\mathbf{Z}}_k$  from the observed inputs  $(\mathbf{Z}_{\leq 0}, \mathbf{a})$  alone, so its error at horizon  $k$  is its per-horizon error  $\delta_k$  and does not depend on its predictions at other horizons.  $\square$

## B.3 PLUG-IN HEADS FOR PUBLISHED WORLD MODELS

For a published backbone with prediction  $\mathbf{B}_k$ , the head outputs  $\hat{\mathbf{Z}}_k = (1 - g_k)\mathbf{B}_k + g_k\mathbf{T}_k$ , where  $\mathbf{T}_k$  is the local  $7 \times 7$  soft transport, with identity bias, of *observed* encoder tokens. The gate  $g_k = \sigma(\mathbf{w}^\top \mathbf{h} + b)$  starts at  $\mathbf{w} = \mathbf{0}$  and  $b = -4$ , so training starts numerically from the unmodified backbone, and nothing else in the backbone, loss or schedule changes.

Table 2: **SHIFTWM as a plug-in head for published world models.** Same data, schedule and loss for each pair; only the head differs. Top: V-JEPA 2-AC on DROID test (its own feature space). Bottom: DINO-WM, official code; teacher-forced latent error on validation; SSIM, LPIPS: quality of decoded frames.

| V-JEPA 2-AC (ViT-g, 1.3B) on DROID               |  | MSE ↓        | moving ↓     | static ↓     | skill (%) ↑ |
|--------------------------------------------------|--|--------------|--------------|--------------|-------------|
| V-JEPA 2-AC, zero-shot                           |  | 0.504        | 0.815        | 0.400        | 4.1         |
| V-JEPA 2-AC, fine-tuned (2 seeds)                |  | 0.396        | 0.741        | 0.281        | 24.6        |
| V-JEPA 2-AC, fine-tuned + SHIFTWM head (2 seeds) |  | <b>0.334</b> | <b>0.616</b> | <b>0.240</b> | <b>36.4</b> |

| Env   | DINO-WM (ViT-S/14)     | latent err. ↓ | SSIM ↑       | LPIPS ↓      |
|-------|------------------------|---------------|--------------|--------------|
| PushT | DINO-WM                | 0.105         | 0.976        | 0.014        |
| PushT | DINO-WM + SHIFTWM head | <b>0.097</b>  | <b>0.977</b> | <b>0.013</b> |
| Wall  | DINO-WM                | <b>0.026</b>  | <b>0.993</b> | <b>0.009</b> |
| Wall  | DINO-WM + SHIFTWM head | 0.031         | 0.988        | 0.026        |

**V-JEPA 2-AC.** Queries come from the predictor’s normalised hidden state for each output block (1024-D), and keys and values from the layer-normalised ViT-g/16 tokens ( $16 \times 16$ , 1408-D) of the last  $S=3$  observed frames, with a learned per-source key offset (0.16M parameters). The fine-tuning objective is the released one (teacher-forced L1 plus rollout L1). During rollouts the predictor consumes  $\hat{\mathbf{Z}}_k$  as its own prediction, while  $\mathbf{T}_k$  only ever reads observed tokens.

**DINO-WM.** We subclass the official `ViTPredictor`, otherwise unmodified. Queries come from the predictor’s output token, and keys and values from its input token at the same slot on the  $14 \times 14$  DINOv2 patch grid ( $S=1$ ); proprioceptive and action channels are left untouched. In rollouts, the input token at that slot is DINO-WM’s own previous prediction after the first step, so the head then moves predicted rather than observed features, unlike SHIFTWM and the V-JEPA 2-AC head. The extra parameters are drawn from a private random generator, so data order, initialisation and dropout of the base predictor are identical in both arms.

## B.4 SOFTWARE

The forecaster, the plug-in head and the evaluation helpers are packaged as a small Python library, `shiftwm` (MIT licence, depending only on PyTorch and NumPy), separate from the research code. It exposes `ShiftWM`, the forecaster of Section 3.2, `ShiftHead`, the backbone-agnostic plug-in head of Appendix B.3 with its gate closed at initialisation, `forecast`, which also returns the transport weights, gate and correction, `transport_field`, the expected source offset of every patch, and `skill`. Its tests confirm that the package reproduces the research implementation (maximum absolute difference 0 on the trained DROID checkpoint) and that `ShiftHead` starts at the backbone’s prediction.

## C EXTENDED EXPERIMENTAL DETAILS

### C.1 DATASETS AND SPLITS

Table 3: **Datasets.** Every forecasting task predicts  $K=10$  ( $K=12$  on IWS) future feature grids from  $H=3$  observed frames and the actions. Step: one model time step. Splits are disjoint at the split unit. DINO-WM PushT and Wall use the official releases and splits.

| Dataset                   | Recorded scenes                                                                                                                                                   | Episodes (train/val/test) | Step        | Action     | Split     | Licence     |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------|------------|-----------|-------------|
| DROID (24 shards)         | real Franka arm in homes, labs and offices; two exterior cameras                                                                                                  | 851 / 143 / 132           | 0.33 s      | 35-D       | session   | CC-BY-4.0   |
| Open-H Hamlyn             | dVRK arms on 7 surgical practice tasks (knot tying, needle grasp and handover, peg transfer, two suturing tasks, tissue lifting, tissue retraction); scene camera | 708 / 152 / 153           | 0.33 s      | 80-D       | episode   | CC-BY-4.0   |
| Language-Table            | xArm pushing coloured blocks; fixed camera (Open-X subset)                                                                                                        | 2,079 / 246 / 264         | 0.1 s       | 2-D        | episode   | CC-BY-4.0   |
| IWS PushT / Box / Rope    | real bimanual T-block, box and rope tasks (Wang et al., 2026b)                                                                                                    | 510–512 / 90 / 200        | 0.17 s      | 20/70/40-D | recording | as released |
| PushT / TwoRoom / Reacher | simulated T-block pushing, two-room navigation and two-link reaching (LeWM planning suites)                                                                       | 2,607 / 290 (PushT)       | 5 env steps | 10-D       | episode   | MIT         |

Table 3 summarises the datasets. **DROID**. We use 24 official full-release shards with 1,126 episodes from 14 sites, split by recording session with a salted hash (316/59/58 sessions in train/val/test). Frames are subsampled from 15 to 3 Hz, each action block concatenates the 5 commanded poses and gripper positions of one step, and the second exterior camera records the same instants from another viewpoint. **Open-H Hamlyn**. The `observation.images.color` stream, which the release labels “endoscope”, is an Intel RealSense scene camera (848×480). Each action block holds five commands, each an 8-D pose-and-gripper target per arm; 6 of 1,019 episodes are shorter than 14 steps and removed, and splits are 70/15/15 per task by a hash of the episode id. **Language-Table**. We use 14 of the 1,024 Open-X shards of a fixed release (6,161 episodes) at the native 10 Hz, since the episodes are short segments of longer play (median 12 steps), and keep the 2,589 episodes long enough for  $H+K=13$  steps, split 81/9/10% (train/val/test) by a salted hash of the episode id. The release has no id of the source play episode, so segments of one play session can fall into different splits; for only 1 of 264 test segments does a training segment start or end with a frame closer to the test boundary frame than the median one-step frame change, but source-level disjointness is not guaranteed. **IWS**. We use the IWS subset of the RLA-WM release<sup>5</sup> (Wang et al., 2026b; Zhang et al., 2026a), real bimanual recordings of pushing a T-block, handling a box and manipulating a rope from one RGB camera at 30 fps (640×480, resized to 224×224). One model step groups five native command rows (1/6 s), so the official 60-row horizon is  $K=12$  steps (2 s), and the action block concatenates the five recorded command vectors (4-D for PushT, 14-D for box, 8-D for rope). The test set is the 200 official validation handles per task, drawn from 10 reserved recordings. Each handle provides a single observed frame, which we replicate into the  $H$ -frame history with neutral past actions. Training and checkpoint selection use the remaining upstream training recordings, split by recording (about 510 training and 90 validation recordings per task). **Planning suites**. We use the PushT, TwoRoom and Reacher expert datasets of the LeWM release<sup>6</sup> with episodes split disjointly into training and validation sets.

## C.2 BASELINES, TRAINING AND COST

**Optional action-contrastive loss.** An optional action-contrastive hinge with margin  $m$  and stop-gradient `sg` makes the forecast depend on the actions by comparing the error under the true future actions with the error under actions borrowed from another sample in the batch,

$$\mathcal{L}_{\text{act}} = \max(0, m - (\|\hat{\mathbf{Z}}(\tilde{\mathbf{a}}) - \mathbf{Z}\|^2 - \text{sg} \|\hat{\mathbf{Z}}(\mathbf{a}) - \mathbf{Z}\|^2)), \quad \mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{act}}. \quad (10)$$

All forecasting and plug-in results use  $\lambda=0$ , like every baseline; the loss enters only as an ablation (Section 5.4).

All learned baselines share the memory encoder, decoder, width, depth, optimiser and schedule of SHIFTWm, and differ only in the output head and rollout. AR is trained through its full 10-step rollout, and AR-TF, following Zhou et al. (2025), regresses absolute next-step features with teacher forcing and is rolled out autoregressively at test time. We evaluate the released V-JEPA 2-AC (ViT-g encoder, 300M predictor) zero-shot on exactly our test windows with its own preprocessing, action convention and rollout, and we fine-tune it on our training split with and without the head under an identical budget; its training data likely include our test trajectories. Validation MSE is computed every 2k steps, and the best checkpoint is tested (Table 4). SHIFTWm has 21.75M parameters against 21.70M for Direct, AR and AR-TF, and because SHIFTWm and Direct decode all  $K$  steps in parallel, a 10-step forecast takes one decoder pass instead of ten. One 8k-step training run takes 0.55 h for SHIFTWm, 0.47 h for Direct, 0.91 h for AR and 0.84 h for AR-TF (median, two runs sharing one H200); the final analyses and planning runs used A100 GPUs.

## C.3 STATISTICAL PROCEDURES

Each test episode contributes the mean of its windows (stride 2), and tables report the mean over training seeds. For a paired comparison of methods A and B, we compute per-episode differences (averaged over seeds) and draw 10,000 bootstrap resamples of episodes, or of recording sessions for DROID, to form a percentile 95% interval. Table daggers use unadjusted intervals; for every dagger of Table 1 no resample crosses zero ( $p < 10^{-4}$ ), so all remain significant after Holm correction

<sup>5</sup> <https://huggingface.co/datasets/xyzhang368/RLA-WM> <sup>6</sup> <https://github.com/lucas-maes/le-wm>; environments and planner from <https://github.com/galilai-group/stable-worldmodel>.

Table 4: **Hyperparameters** (shared by all learned predictors unless noted).

|                                       |                       |                     |                                |
|---------------------------------------|-----------------------|---------------------|--------------------------------|
| Width / heads / MLP ratio             | 384 / 6 / 4           | Optimiser           | AdamW, $\beta=(0.9, 0.95)$     |
| Memory encoder layers $L_e$           | 4                     | Peak LR / warm-up   | $3 \times 10^{-4}$ / 800 steps |
| Query decoder layers $L_d$            | 4                     | Schedule            | cosine, 16k steps              |
| History $H$ / horizon $K$             | 3 / 10                | Batch size          | 48 windows                     |
| Transport window $w$ / sources $S$    | 7 / 3                 | Weight decay / clip | 0.05 / 1.0                     |
| Key dim / identity bias init          | 64 / 4.0              | Precision           | bf16 autocast                  |
| Gate bias init                        | -3                    | Selection           | best val. MSE (every 2k steps) |
| Action-contrastive $\lambda$ / margin | 1.0 / 0.05 (planning) | Seeds               | 0, 1, 2                        |

over the three learned baselines of each dataset. Training seeds per method are 3 for DROID, 2 for V-JEPA 2-AC and 1 for Open-H, Language-Table, IWS and the ablations; on DROID the full model’s seed range is 0.2% of its validation error.

## D EXTENDED RESULTS AND ANALYSIS

### D.1 FULL FORECASTING RESULTS

Figure 10 breaks the results of Table 1 down by horizon and by surgical task. Every learned predictor lies below persistence except AR-TF on DROID beyond  $k=4$ , and SHIFTFWM is lowest at every horizon and on all seven surgical tasks. Table 1 averages Hamlyn over episodes, weighting tasks by their number of test episodes, and Table 5 gives the three IWS tasks, where SHIFTFWM has the lowest average error on each, significantly on Box and Rope.

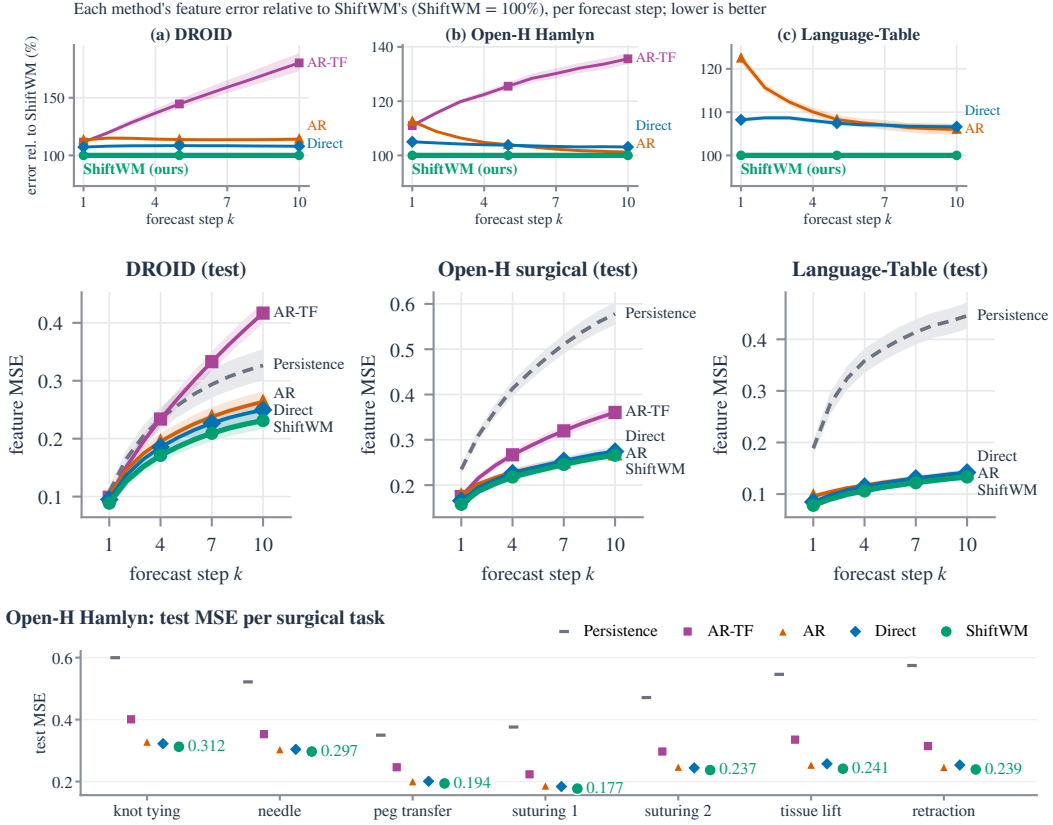


Figure 10: **Forecast error per step  $k$  and per surgical task.** Top: each baseline's error relative to SHIFTFWM's (SHIFTFWM = 100%; bands: paired 95% bootstrap). Middle: absolute feature MSE. Bottom: Hamlyn test MSE per task (mean over horizons and 1 seed(s)).

Table 5: **IWS per task** (feature MSE  $\downarrow$  on the 200 official validation handles per task, seed 0,  $K=12$ ). **Best**, second;  $\dagger$ : paired 95% CI (handles) vs. the best baseline excludes 0.

| Method                                     | PushT             |              | Box                    |              | Rope                   |                        |
|--------------------------------------------|-------------------|--------------|------------------------|--------------|------------------------|------------------------|
|                                            | avg               | $k=12$       | avg                    | $k=12$       | avg                    | $k=12$                 |
| Persistence                                | 0.738             | 0.865        | 0.735                  | 0.885        | 0.749                  | 0.863                  |
| AR-TF (DINO-WM-style)                      | 0.269             | 0.306        | 0.305                  | 0.347        | 0.217                  | 0.234                  |
| AR (rollout-trained)                       | 0.238             | <b>0.255</b> | 0.256                  | <b>0.272</b> | 0.204                  | 0.212                  |
| Direct (cross-attn)                        | 0.245             | 0.273        | 0.256                  | 0.279        | 0.198                  | 0.208                  |
| SHIFTFWM (ours)                            | <b>0.236</b>      | 0.268        | <b>0.250</b> $\dagger$ | 0.276        | <b>0.189</b> $\dagger$ | <b>0.198</b> $\dagger$ |
| error reduction vs. best baseline [95% CI] |                   |              |                        |              |                        |                        |
|                                            | +0.7% [-0.2, 1.7] |              | +2.4% [1.7, 3.2]       |              | +5.0% [4.7, 5.2]       |                        |

**Action ranking.** A forecast that stays close to  $\mathbf{Z}_0$  can score well on average while barely responding to the actions, and a planner then cannot rank candidate action sequences. We therefore measure the

Table 6: **Error by region and decoded-pixel quality.** Left: DROID test feature MSE ↓ on moving and static patches and overall; skill = error removed vs. persistence. Mean over 3 training seeds, as in Table 1. Right: forecasts at  $k=10$  decoded to RGB by one decoder per dataset, shared by all methods; the first row decodes the true features.

| Method                   | DROID feature error by region |              |              |              | DROID decoded |       |        | Open-H Hamlyn decoded |              |        |
|--------------------------|-------------------------------|--------------|--------------|--------------|---------------|-------|--------|-----------------------|--------------|--------|
|                          | moving                        | static       | all          | skill ↑      | PSNR↑         | SSIM↑ | LPIPS↓ | PSNR↑                 | SSIM↑        | LPIPS↓ |
| Decoder on true features | –                             | –            | –            | –            | 18.15         | 0.548 | 0.196  | 32.72                 | 0.931        | 0.030  |
| Persistence              | 0.726                         | 0.090        | 0.249        | 0.0%         | 16.05         | 0.498 | 0.256  | 21.54                 | 0.832        | 0.143  |
| AR-TF (DINO-WM-style)    | 0.582                         | 0.173        | 0.275        | -10.4%       | 15.32         | 0.465 | 0.299  | 24.92                 | 0.862        | 0.093  |
| AR (rollout-trained)     | 0.513                         | 0.103        | 0.206        | 17.4%        | 16.15         | 0.487 | 0.262  | 27.54                 | 0.892        | 0.065  |
| Direct (cross-attn)      | 0.488                         | 0.098        | 0.196        | 21.6%        | 16.30         | 0.492 | 0.253  | 27.31                 | 0.891        | 0.067  |
| SHIFTWM (ours)           | <b>0.453</b>                  | <b>0.090</b> | <b>0.181</b> | <b>27.5%</b> | <b>16.40</b>  | 0.495 | 0.254  | <b>27.60</b>          | <b>0.893</b> | 0.066  |

*action-ranking error*, the share of test windows in which another episode’s actions yield a lower forecast error than the true actions. On DROID, SHIFTWM has both the highest skill (27.5%) and the lowest ranking error (8.0%, against 10.2% for Direct, 19.0% for AR and 15.1% for AR-TF), as Figure 11 shows. On Hamlyn SHIFTWM is again best on both axes (ranking error 0.55% against 0.64% for Direct), and on Language-Table it has the highest skill while ranking errors are rare for all methods (1.93% against 1.80% for Direct), with overlapping intervals on both. The gain over Direct is positive in every quintile of episode motion on all three datasets (Figure 11d).

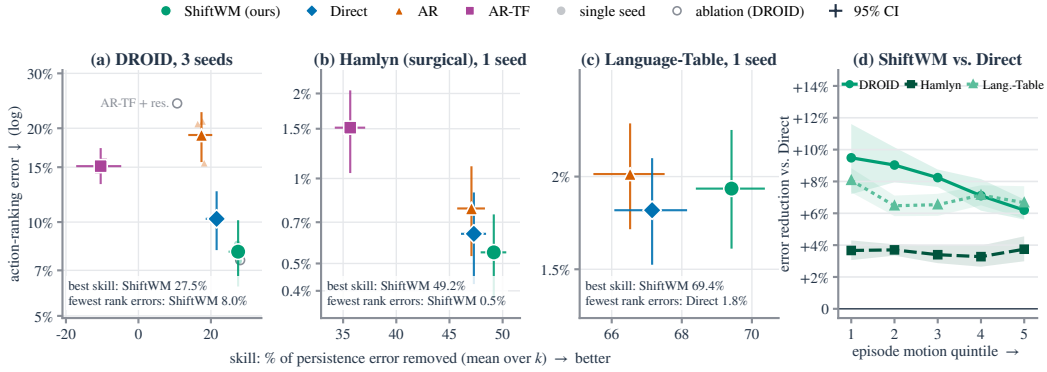


Figure 11: **Forecast skill versus action ranking (test).** (a–c) Lower right is better; large markers: seed means with 95% CIs; hollow: DROID ablations. (d) Error reduction of SHIFTWM over Direct per quintile of episode motion.

## D.2 ERROR BY REGION AND DECODED PIXELS

Table 6 gives the moving/static split of DROID test error behind Section 5.3. Because feature error has no direct visual meaning, we also decode every forecast to RGB with one feature-to-RGB decoder per dataset, trained on true training features (L1 + LPIPS) and applied unchanged to every method. SHIFTWM has the best PSNR on both datasets and the best SSIM on Hamlyn, whereas on DROID persistence has the highest SSIM and Direct the lowest LPIPS.

## D.3 PLUG-IN TRAINING DYNAMICS

Figure 12 shows how the plug-in results of Section 5.2 arise during training, for every plug-in run we trained under an identical budget per pair. For V-JEPA 2-AC (two seeds, 3,000 steps), validation error with the head is lower at every evaluation step, and the gate opens from 0.018 to about 0.55 within the first 250 steps and rises to about 0.65. The gain holds at every horizon, with test error 15% lower at  $k=1$  and 17% lower at  $k=10$ . On DINO-WM PushT (3 epochs), the head lowers teacher-forced latent error by 8% and open-loop error by 6% at the first rollout step and 11% at the fifth. On Wall (15 epochs), teacher-forced error is 18% higher with the head and open-loop error 23–39% higher; the head’s validation error is still falling at the end of the budget. One official DINO-

WM planning run takes about 7 GPU-hours, and within our budget only the unmodified DINO-WM finished on PushT (62.0% success, panel h).

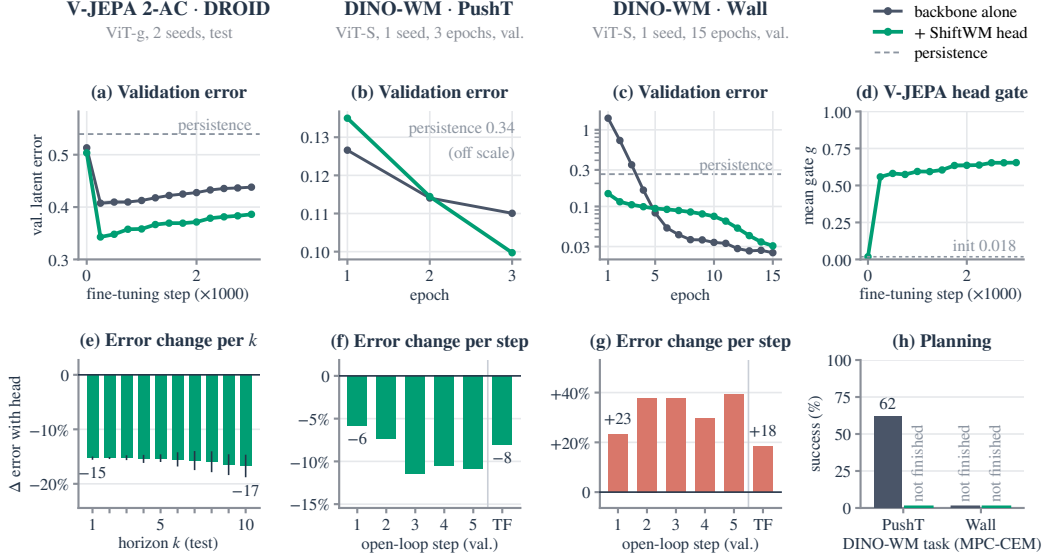


Figure 12: **Plug-in heads during training**, with (green) and without (dark grey) the SHIFTWM head. (a–c) Validation latent error. (d) Mean V-JEPA gate. (e–g) Change in latent error from the head. (h) DINO-WM planning success.

#### D.4 CLOSED-LOOP PLANNING

We evaluate closed-loop planning with the official LeWM and stable-worldmodel protocol (Table 7). Every predictor, including the released LeWM checkpoint, is wrapped in one interface, which reproduces the native LeWM planning costs exactly, and is driven by the same CEM planner, goal sampling, budget and success criteria, so only the world model changes. Among the matched predictors, which share a frozen DINOv2 encoder and a short training budget, PushT success ranges from 58.7% (SHIFTWM) to 78.7% (AR), and the released LeWM, trained end to end, reaches 88% (Figure 13). Lower forecast error thus does not by itself raise planning success here. Forecast error bounds planning regret only along directions that change the ranking of candidate plans. If the planner scores  $\hat{c}(\mathbf{a}) = \|\hat{\mathbf{z}}_T(\mathbf{a}) - \mathbf{z}_g\|^2$  with  $\|\hat{\mathbf{z}}_T - \mathbf{z}_T\| \leq \varepsilon$  and  $\|\mathbf{z}_T - \mathbf{z}_g\| \leq R$ , then  $|\hat{c} - c| \leq 2R\varepsilon + \varepsilon^2$ , and a plan within  $\eta$  of the best predicted cost has true cost at most  $\min_{\mathbf{a}} c(\mathbf{a}) + 4R\varepsilon + 2\varepsilon^2 + \eta$ .

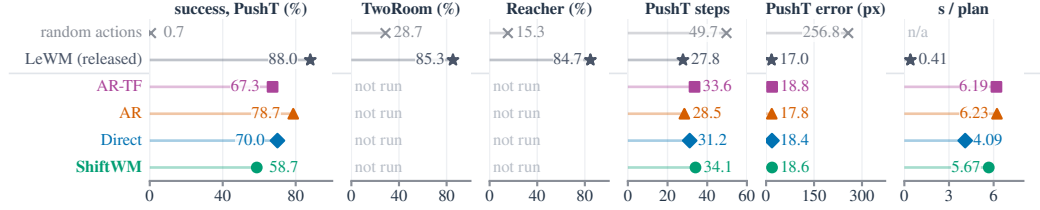
Table 7: **Official planning protocol** (le-wm@8edfeb33, stable-worldmodel@4821c8e6).

|                      |                                                                                      |
|----------------------|--------------------------------------------------------------------------------------|
| Planner              | CEM, 300 samples × 30 iterations, 30 elites, initial variance 1, warm start          |
| Horizon / replanning | 5 action blocks of 5 env steps; replan every 25 steps                                |
| Episodes / seeds     | 50 episodes per planner seed; seeds 42, 43, 44                                       |
| Goal                 | dataset frame 25 steps later in the same trajectory                                  |
| Budget               | 50 environment steps                                                                 |
| Cost                 | squared distance between final predicted latent and goal latent                      |
| Success              | PushT: position < 20 px and angle < $\pi/9$ ; TwoRoom: < 16 px; Reacher: joint match |

**Reproduction check.** With the released LeWM PushT checkpoint, our harness reaches 45/50, 44/50 and 43/50 successes for planner seeds 42/43/44 (mean 88%) with the single released training seed, whereas the LeWM paper reports  $96 \pm 2.8$  averaged over training seeds. On TwoRoom and Reacher the released checkpoints reach 85.3% and 84.7%, against 87 and 86 reported by Maes et al. (2026b). LeWM’s own evaluation code gives 46/50 on PushT and 39/50 on Reacher for the same checkpoint and seed 42, against 45/50 and 42/50 in our harness, a difference within planner-seed variation (41–44 in our Reacher runs). Uniform random actions succeed in 0.7% (PushT), 28.7% (TwoRoom) and 15.3% (Reacher) of episodes, so some TwoRoom and Reacher goals lie within reach of undirected

motion. Figure 13 also gives the mean number of steps to the first success, with failures counted at the 50-step budget, and the median final distance to the goal in the units of each success test (PushT position error in px, TwoRoom distance in px, Reacher largest joint-angle error in rad).

success  $\uparrow$  (50 episodes  $\times$  3 planner seeds); steps to first success, final goal error, time per plan  $\downarrow$



References, steps to success and final error: LeWM TwoRoom: 22.3 steps, 14.7 px; LeWM Reacher: 32.3 steps, 0.043 rad; random TwoRoom: 39.5 steps, 41.9 px; random Reacher: 45.4 steps, 0.968 rad

Figure 13: **Closed-loop planning**, official LeWM protocol (50 episodes  $\times$  3 planner seeds). LeWM and random actions are references; matched predictors are planned on PushT only. Steps: mean to first success (budget 50); error: median final distance to the goal.

## D.5 THE MECHANISM ON A CONTROLLED TOY WORLD

To make each step of SHIFTWM visible, we train the same model class on a toy world whose motion is known exactly (Figure 14). Frames of  $64 \times 64$  pixels show a striped background, a red square moved by the 2-D action (4 px per step), a drifting blue disc and an occluder that slides away to reveal a never-observed yellow cross. The features are the raw pixels of each  $4 \times 4$  patch ( $16 \times 16$  grid, 48-D), and a small instance of the architecture ( $H=3$ ,  $K=5$ ,  $11 \times 11$  window) is trained for SHIFTWM, Direct and AR with identical budgets on 1,500 episodes and tested on 300. On the whole test set, the transport recovers the true motion of the square, as its expected source offset is off by 2.8–3.0 px at every step, whereas assuming no motion is off by 4.1 px at  $k=1$  and 14.4 px at  $k=5$ . The gate opens on moving content (0.99 on the square at every step) and stays mostly closed on the background (0.11–0.37), and the correction concentrates on the revealed cross (norm about 1.45 against 0.04 on the background). SHIFTWM has the lowest test error at every step ( $4.8 \rightarrow 13.1 \times 10^{-3}$  over  $k=1..5$ , against Direct  $5.3 \rightarrow 13.3$ , AR  $8.1 \rightarrow 14.9$  and persistence  $18.4 \rightarrow 56.8$ ); on this simple scene Direct is close to SHIFTWM. The window in Figure 14 is selected by a fixed rule, as all three objects are visible and apart in every frame, the square moves at every observed step, the cross is 30–80% revealed at  $t$  and fully revealed at  $t+K$ , and the square’s displacement is the largest among such windows.

## D.6 IS THE FUTURE A TRANSPORT?

The *oracle move* picks, for each moving patch, the observed candidate in its transport window closest to the true future; it uses the future to choose and is therefore a model-free reference, not a predictor. Gain recovered is  $(\text{err}_{\text{stay}} - \text{err}_{\text{model}}) / (\text{err}_{\text{stay}} - \text{err}_{\text{oracle}})$ . Figure 15a shows the  $Sw^2=147$  observed candidates of three patches in a 2-D PCA, sized by the learned transport weight. Among moving, high-gate ( $g>0.5$ ) patches with above-median change, these are the three, from different episodes, with the largest advantage of SHIFTWM over the better baseline, so they are selected and not typical. Panels (b,c) aggregate over all moving patches of 3,923 windows from 130 test episodes, and the paired intervals of the moving-patch error difference at  $k=10$  exclude zero against both Direct and AR.

In Figure 16a,b, every patch at every horizon is assigned to a decile of its true change, and each cell reports  $1 - \text{err}_{\text{SHIFTWM}} / \text{err}_{\text{baseline}}$ . Against Direct the gain is largest for moderate-to-large motion at long horizons, and against AR it is largest on nearly static patches, where the recursive rollout accumulates error on content that should not change. On moving patches the mean gate rises from 0.51 at  $k=1$  to 0.85 at  $k=10$ , and the moved term  $g(\mathbf{T} - \mathbf{Z}_0)$  carries 87% of the squared departure from  $\mathbf{Z}_0$ , so the correction remains a small residual. Per episode, SHIFTWM loses to Direct and to AR on only 3 of 130 episodes each, all in the less-moving half and none in all three seeds. The

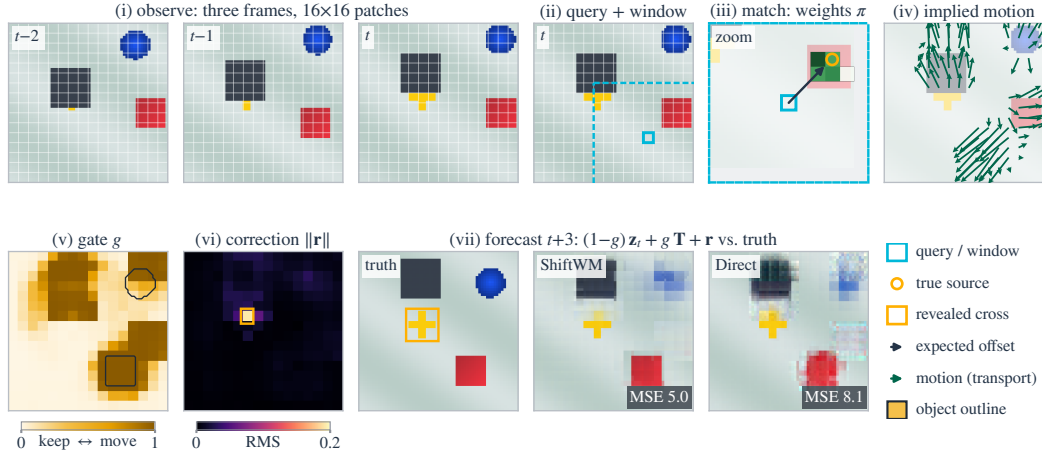


Figure 14: **The mechanism on one toy window.** (i) Observed frames; (ii) a query patch and its window; (iii) transport weights; (iv) motion where the gate exceeds 0.5; (v) gate; (vi) correction; (vii) forecast at  $t+3$ .

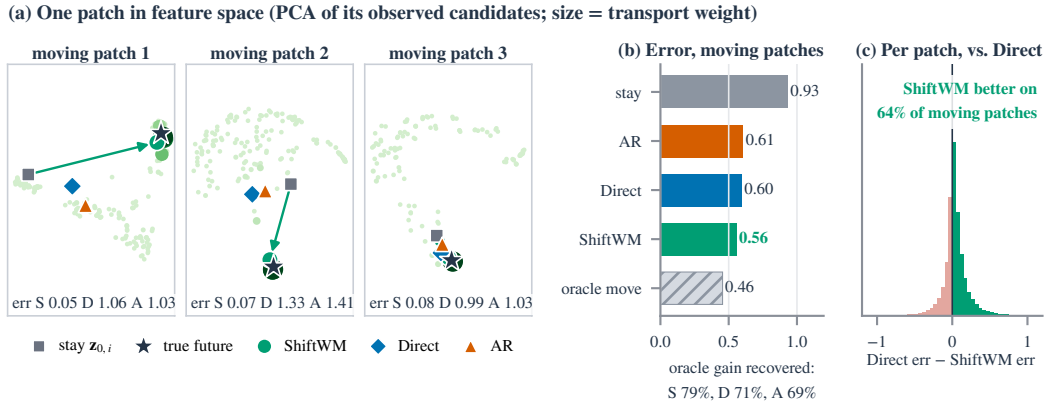


Figure 15: **Feature-space geometry on held-out DROID ( $k=10$ ).** (a) Three selected moving patches in a 2-D PCA of their observed candidates. (b) Moving-patch error with the oracle move. (c) Per-patch error difference to Direct.

relative gain shrinks with episode motion (Spearman  $\rho = 0.38$  between motion and the signed relative difference to Direct), whereas the absolute error reduction grows with it (Spearman  $+0.37$ ; per motion quartile 0.011, 0.014, 0.016 and 0.018), because the baselines' error rises faster than the gain.

## D.7 BEYOND FEATURE ERROR

**Feature warping with optical flow.** We warp the last observed grid  $Z_0$  with RAFT (Teed & Deng, 2020) flow pooled to the patch grid, on the same 3,923 DROID test windows and metric as Table 1. *Flow extrapolation*, in the style of F2M (Šarić et al., 2020), scales the flow between the last two observed frames by  $k$ , whereas the *oracle warp* uses the flow from the true future frame and thus upper-bounds a pure warp of  $Z_0$ . Averaged over horizons, persistence, flow extrapolation, the oracle warp and SHIFTM reach 0.249, 0.242, 0.221 and 0.181. SHIFTM beats even the oracle warp, as its gate keeps static content, where estimated flow is noisy, and its correction adds content that no warp of  $Z_0$  contains.

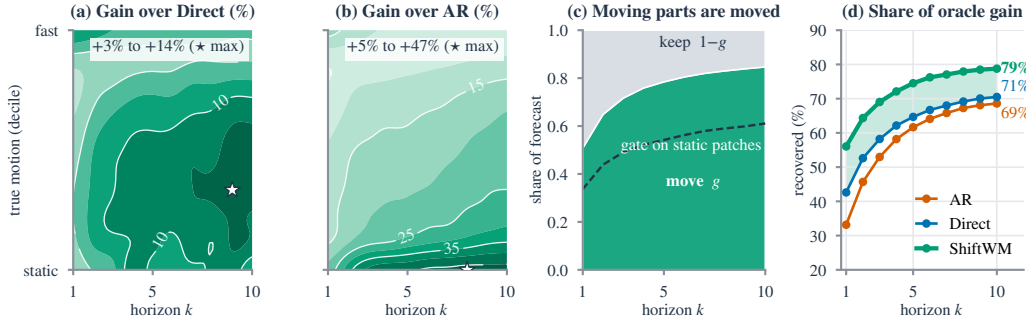


Figure 16: **Anatomy of the gains (held-out DROID).** (a,b) Error reduction over horizon and motion decile. (c) Gate on moving and static patches. (d) Share of the oracle-move gain recovered on moving patches.

**Segmentation view.** Feature error does not show *where* a forecast puts the arm, so we read a segmentation out of every forecast (Figure 17). Grounding DINO (Liu et al., 2024)<sup>7</sup> detects the arm or the instruments on the last observed frame and SAM 2.1 (Ravi et al., 2025)<sup>8</sup> tracks the mask through the true future; windows with a confident, stable track covering 3–128 of the 256 patches are kept (1,332 of 1,500 DROID and 1,017 of 1,500 Hamlyn windows). A forecast patch is foreground if it is more similar (cosine) to a foreground than to a background patch of  $\mathbf{Z}_0$  (IoU 0.70 on true DROID features at  $k=10$ ); moving windows have below-median mask overlap between  $t$  and  $t+K$ , and placement is the centroid distance. On DROID, SHIFTWM has the highest IoU at every reported horizon, on all and on moving windows, with paired intervals that exclude zero against both Direct and AR, and it has the lowest placement error on moving windows (12.0 px against 12.2 for Direct and 13.9 for AR). On Hamlyn, where the instruments move less within ten steps, the three learned predictors are within half an IoU point of each other.

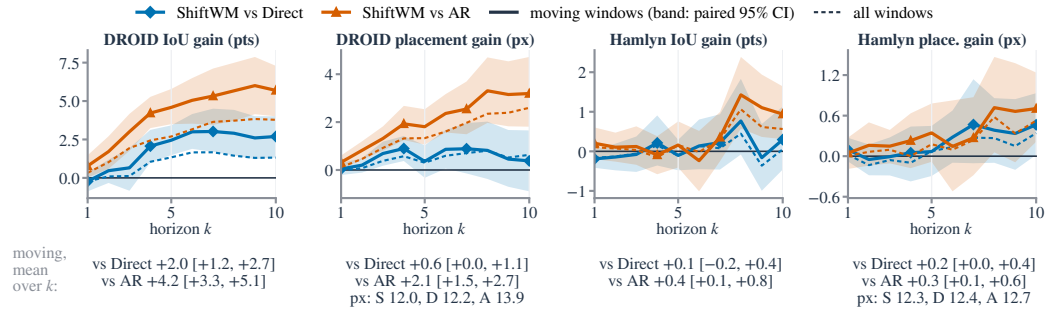


Figure 17: **Segmentation read out of the forecasts.** Paired gain of SHIFTWM over Direct and AR per horizon: IoU (points) and placement error removed (px of the displayed frame); bands: paired 95% CIs on moving windows. Below: moving-window means over  $k$  with CIs, and mean placement error of SHIFTWM (S), Direct (D) and AR (A).

## D.8 QUALITATIVE RESULTS

**Example selection.** Every method is shown on the same window. The teaser, the bottom row of Figure 2 and Figure 5a use held-out DROID windows among the 30% with the most motion that have the largest step-10 advantage of SHIFTWM over the better of Direct and AR (over Direct for Figure 5a); typical behaviour is given by the aggregate results, as SHIFTWM has lower error than

<sup>7</sup> <https://github.com/IDEA-Research/GroundingDINO>; both models via Hugging Face transformers.

<sup>8</sup> <https://github.com/facebookresearch/sam2>

Direct on 98% of DROID test episodes (Figure 6). In Figure 1a, step-10 features are decoded to RGB by the decoder of Appendix D.2; error maps share one scale, dashed outlines mark moving patches (the 25% with the largest true change), and arrows show the transport of patches with  $g \geq 0.5$  that move by at least half a patch. Figure 1b uses the region errors of Table 6; Figure 1c compares with Direct on DROID and Hamlyn and with the model without the head for V-JEPA 2-AC and DINO-WM. Figure 18 selects, among stride-2 test windows in the top half of true motion, the three DROID windows (distinct episodes) and the Hamlyn window with the largest relative  $k=10$  gain  $1 - e_{\text{SHIFTWM}} / \min(e_{\text{Direct}}, e_{\text{AR}})$  and, as the matched failure, the DROID window with the largest SHIFTWM error, where the three models err alike (within 7%; AR lowest).

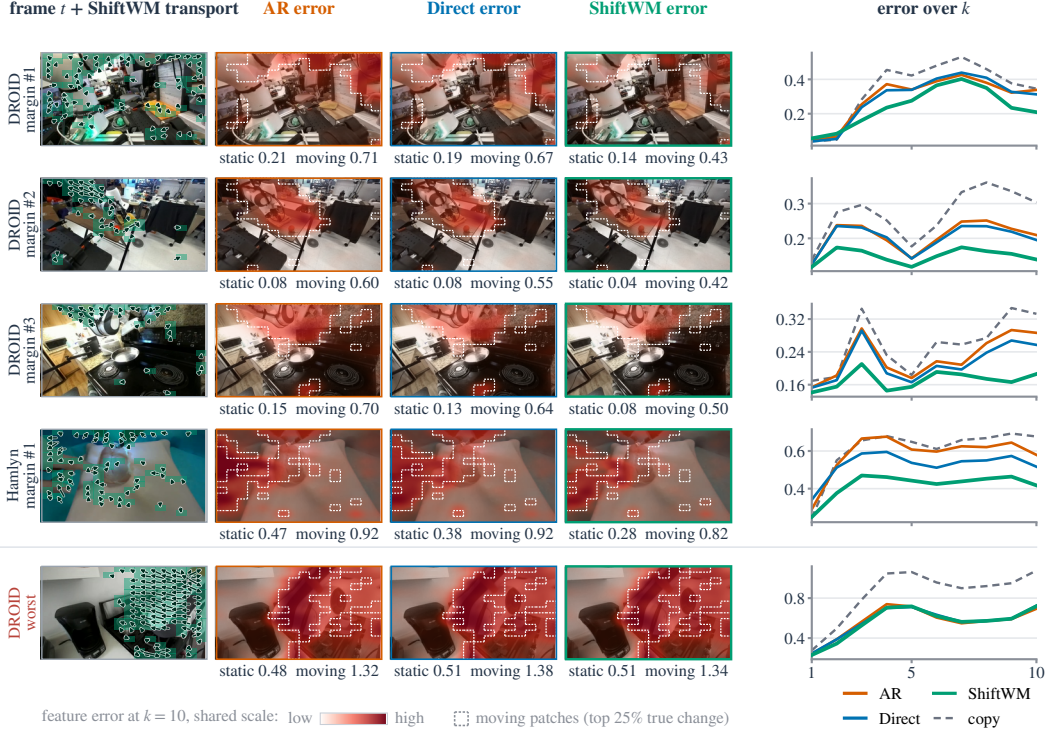


Figure 18: **Largest-margin and worst test windows.** Top: DROID and Hamlyn windows (top half of motion) with the largest  $k=10$  gain of SHIFTWM over the better of Direct and AR; bottom: SHIFTWM’s worst DROID window. Error maps share one scale (dashed: moving patches). SHIFTWM beats Direct on 98% of DROID test episodes (Figure 6).

## E DISCUSSION

**Scope.** Transport helps most where content moves within the window and keeps its appearance; content that no observed frame contains relies on the correction, and longer histories or adaptive windows are natural extensions.

**Uncertainty.** Under uncertain motion, the MSE-optimal forecast averages over plausible sources (Proposition 1). SHIFTWM averages observed features rather than regressing a mean, but it remains a deterministic model, and stochastic feature forecasters (Boduljak et al., 2025; Porcher et al., 2026) that target multimodal futures are complementary.